

ROBINSON, IL, USA

WMO: 720319

Lat: 39.016N

Lon: 87.650W

Elev: 462

StdP: 14.45

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 63841

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 4.3	(c) 9.8	(d) -3.6	(e) 4.6	(f) 6.4	(g) 1.2	(h) 5.9	(i) 11.6	(j) 26.2	(k) 41.1	(l) 24.3	(m) 36.7	(n) 5.3	(o) 310	(p) 0.467

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.3	92.8	75.5	90.6	74.9	88.4	73.9	79.8	88.3	78.2	86.3	76.8	84.5	7.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.4	144.7	84.6	75.8	137.3	83.0	74.4	130.8	81.4	43.5	88.4	42.0	85.6	40.4	84.8	91.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 23.0	(b) 19.5	(c) 17.3	(e) -3.4	(f) 95.9	(g) 6.2	(h) 2.9	(i) -7.8	(j) 97.9	(k) -11.5	(l) 99.6	(m) -14.9	(n) 101.3	(o) -19.4	(p) 103.4		
(4) 23.0	(5) 19.5	(6) 17.3	(7) -3.4	(8) 95.9	(9) 6.2	(10) 2.9	(11) -7.8	(12) 97.9	(13) -11.5	(14) 99.6	(15) -14.9	(16) 101.3	(17) -19.4	(18) 103.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.9	30.3	33.0	44.8	55.5	65.9	74.2	75.5	73.9	68.4	56.5	44.0	35.1
	DBStd	18.50	11.49	11.83	11.24	9.27	8.51	5.65	5.23	5.21	6.55	9.11	9.44	10.05
	HDD50	2124	616	487	238	49	2	0	0	0	0	35	226	471
	HDD65	4904	1075	895	630	303	97	4	1	5	40	296	631	927
	CDD50	3908	6	12	77	215	497	726	790	741	551	237	46	10
	CDD65	1213	0	0	5	20	126	281	326	281	141	33	0	0
	CDH74	11975	0	0	41	218	1181	2707	3242	2711	1516	353	6	0
	CDH80	4525	0	0	4	35	385	1083	1318	1035	563	102	0	0
Wind	WSAvg	6.9	9.1	8.7	8.8	8.8	7.1	5.9	4.4	3.8	4.3	6.4	7.4	8.0
Precipitation	PrecAvg	41.0	2.6	2.3	3.7	3.9	4.5	3.6	4.1	3.5	3.2	2.9	3.6	3.0
	PrecMax	55.0	9.4	5.6	9.4	9.6	10.4	6.3	10.8	7.8	9.0	8.4	11.4	8.1
	PrecMin	28.5	0.2	0.4	0.7	1.1	1.2	0.7	0.8	0.9	0.1	0.5	0.4	0.2
	PrecStd	7.0	2.2	1.2	2.0	1.9	2.5	1.6	2.2	1.9	2.4	1.9	2.0	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.2	67.7	79.4	84.3	91.0	95.0	99.7	95.1	93.0	88.1	74.4	65.5
		MCWB	58.2	58.6	62.4	64.3	71.2	74.2	75.2	76.9	71.6	70.3	59.5	60.8
	2%	DB	57.3	61.9	73.0	79.5	87.8	91.7	92.6	91.3	89.9	82.3	69.0	59.6
		MCWB	53.2	56.5	60.6	63.5	69.3	75.0	76.8	75.8	72.6	67.0	60.5	56.4
	5%	DB	52.4	57.1	67.7	76.2	84.5	89.7	90.0	88.9	86.4	77.4	63.9	54.8
		MCWB	49.1	52.0	58.2	61.9	68.9	73.6	77.1	75.4	72.0	64.3	56.7	51.4
	10%	DB	47.2	50.8	63.3	72.1	81.2	86.5	87.2	86.0	82.9	72.7	59.7	50.3
		MCWB	44.0	45.7	56.4	60.4	67.5	71.8	75.5	74.3	69.5	62.3	52.3	47.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.9	61.3	64.9	69.6	74.9	80.1	82.1	82.1	78.1	73.2	65.7	62.1
		MCDB	62.1	65.6	74.1	77.2	84.9	90.1	89.7	90.8	85.7	84.5	71.2	64.8
	2%	WB	54.5	57.5	62.6	66.6	73.0	78.0	80.1	79.4	76.3	70.6	61.5	56.9
		MCDB	57.0	61.1	70.6	74.0	82.7	87.5	88.2	87.3	84.5	78.8	66.0	59.1
	5%	WB	49.8	52.8	60.0	64.3	71.4	76.2	78.4	77.7	74.1	67.8	57.9	52.3
		MCDB	51.9	56.5	66.9	71.8	80.3	85.1	86.3	85.4	82.5	73.7	63.4	54.3
	10%	WB	43.4	45.9	56.8	62.2	69.7	74.5	77.0	76.0	72.1	64.6	53.5	47.0
		MCDB	46.4	50.5	61.9	69.9	78.0	82.7	84.5	82.7	79.6	70.8	57.9	49.6
Mean Daily Temperature Range	5% DB	MDBR	15.8	17.0	18.9	21.4	20.2	20.5	20.3	21.8	24.8	22.9	19.2	14.7
		MCDBR	21.3	24.1	25.0	26.9	24.3	23.8	22.8	24.5	28.3	28.5	25.6	20.1
		MCWBR	18.0	19.4	15.0	13.9	9.9	9.1	9.6	11.4	12.8	14.2	16.8	17.3
	5% WB	MCDBR	19.6	22.7	22.6	21.3	20.9	20.0	19.2	20.0	22.7	23.1	22.4	18.7
		MCWBR	18.1	19.7	14.5	13.4	10.1	8.8	9.8	11.1	11.9	13.0	17.3	17.5
Clear-Sky Solar Irradiance	taub	0.310	0.321	0.357	0.408	0.444	0.448	0.467	0.476	0.426	0.363	0.322	0.305	
	taud	2.445	2.399	2.371	2.229	2.180	2.203	2.150	2.110	2.236	2.413	2.477	2.470	
	Ebn at Noon	274	286	285	276	267	265	259	252	259	267	267	267	
	Edh at Noon	26	31	35	43	47	45	48	48	40	30	25	23	
All-Sky Solar Radiation	RadAvg	590	842	1174	1535	1737	2000	1938	1768	1472	1051	706	504	
	RadStd	67	93	103	95	153	125	121	105	105	109	75	52	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+185	N/A

Nomenclature: See separate page